

Briefly describe the artifact. What is it? When was it created?

The artifact is an Android inventory management application that I created in CS 360 Mobile Architecture and Programming. It was developed as the final project for that course. The app was built in Java using Android Studio and allows users to create accounts, log in, and manage inventory items using create, read, update, and delete functions. The app stores data using a SQLite database and includes an alert feature when an item reaches a quantity of zero.

Justify the inclusion of the artifact in your ePortfolio. Why did you select it? What specific components showcase your skills? How was it improved?

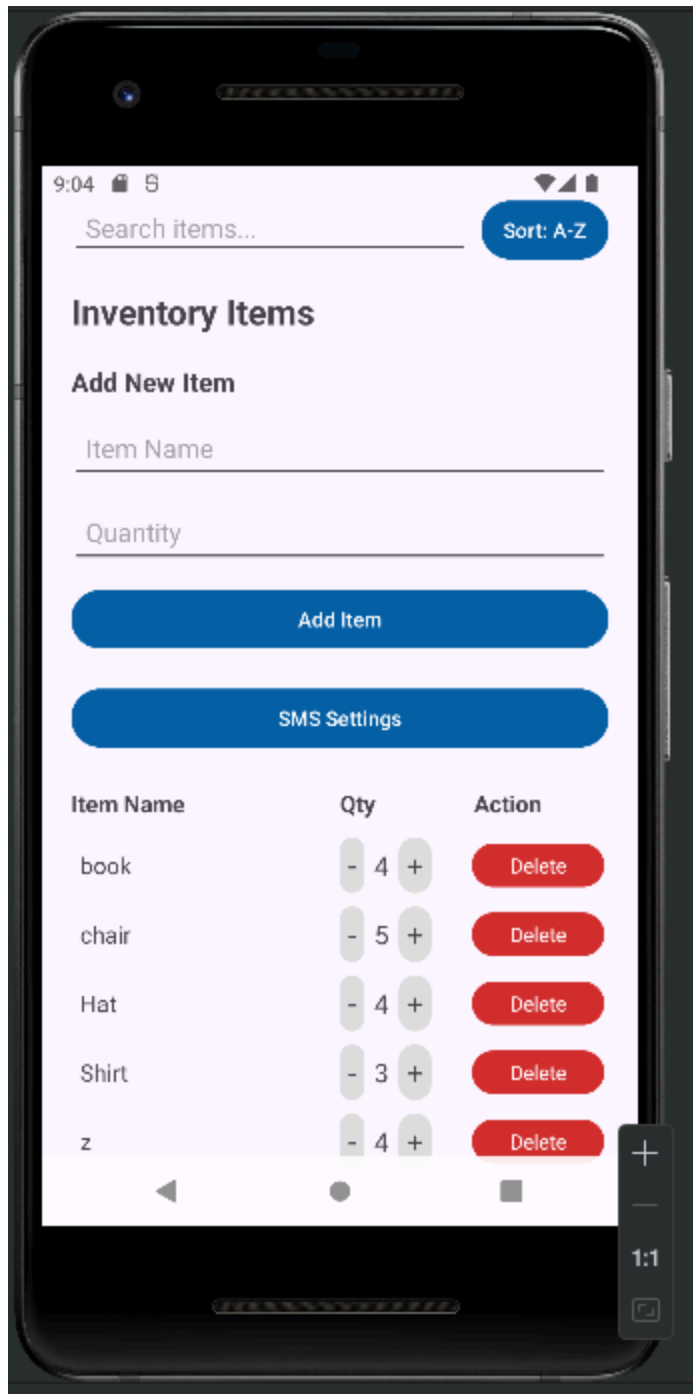
I selected this artifact for the algorithms and data structures category because the original version had no way to search or sort inventory items. Users had to manually scroll through the entire list to find what they were looking for, which becomes a real problem as the inventory grows larger. The method that loads the inventory list was also rebuilding the entire table from scratch every time any button was clicked, even for small changes like updating a quantity.

The main improvement in this enhancement is the addition of a real time search bar and a sort toggle button to the inventory screen. As the user types into the search bar the list filters instantly to show only items that match the search input. The sort button toggles between two modes, sorting items alphabetically from A to Z and sorting by lowest quantity first. Sorting by quantity is especially useful for quickly identifying which items are running low.

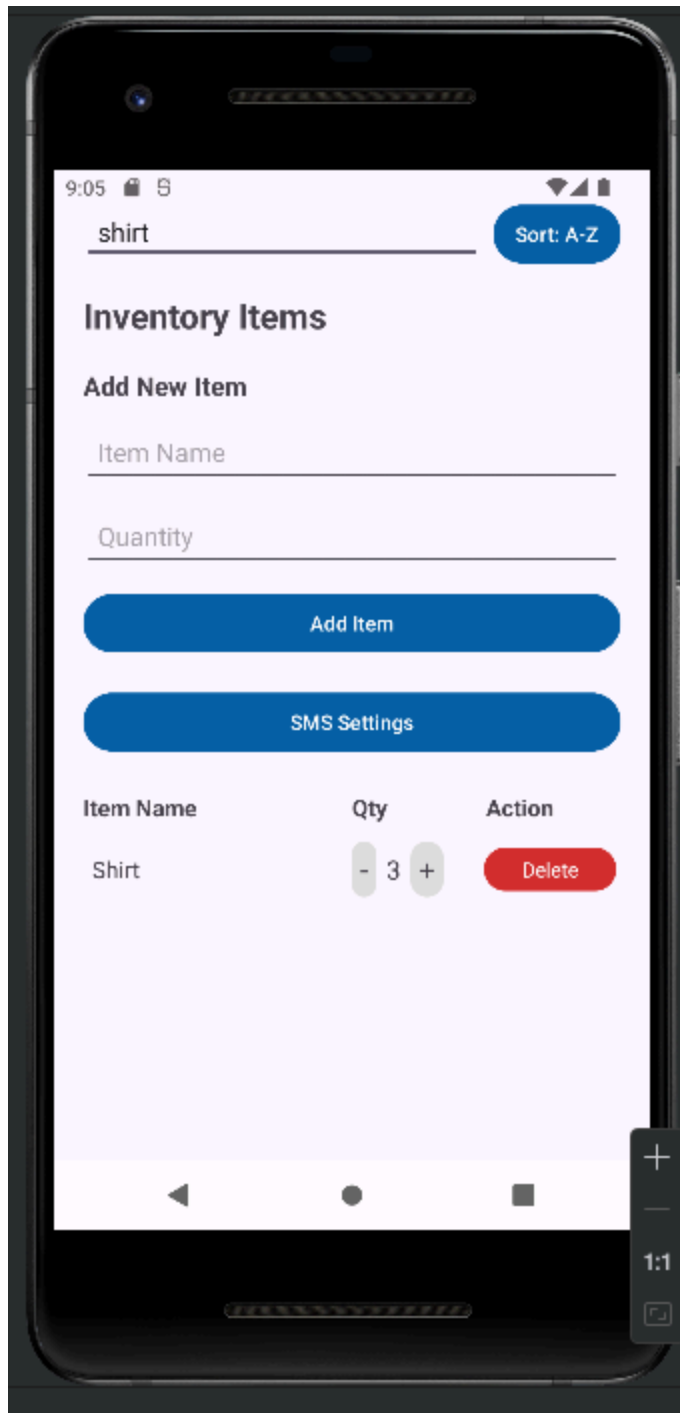
The search was implemented using a SQL query at the database level rather than loading all items into memory and filtering them in Java. This is a more efficient approach because the database handles the filtering directly, which keeps the app performing well even as the inventory grows. The sort uses an ORDER BY clause in the same query so both search and sort work together at the same time.

Below are screenshots of the enhanced inventory application that demonstrate the improvements made during this milestone.

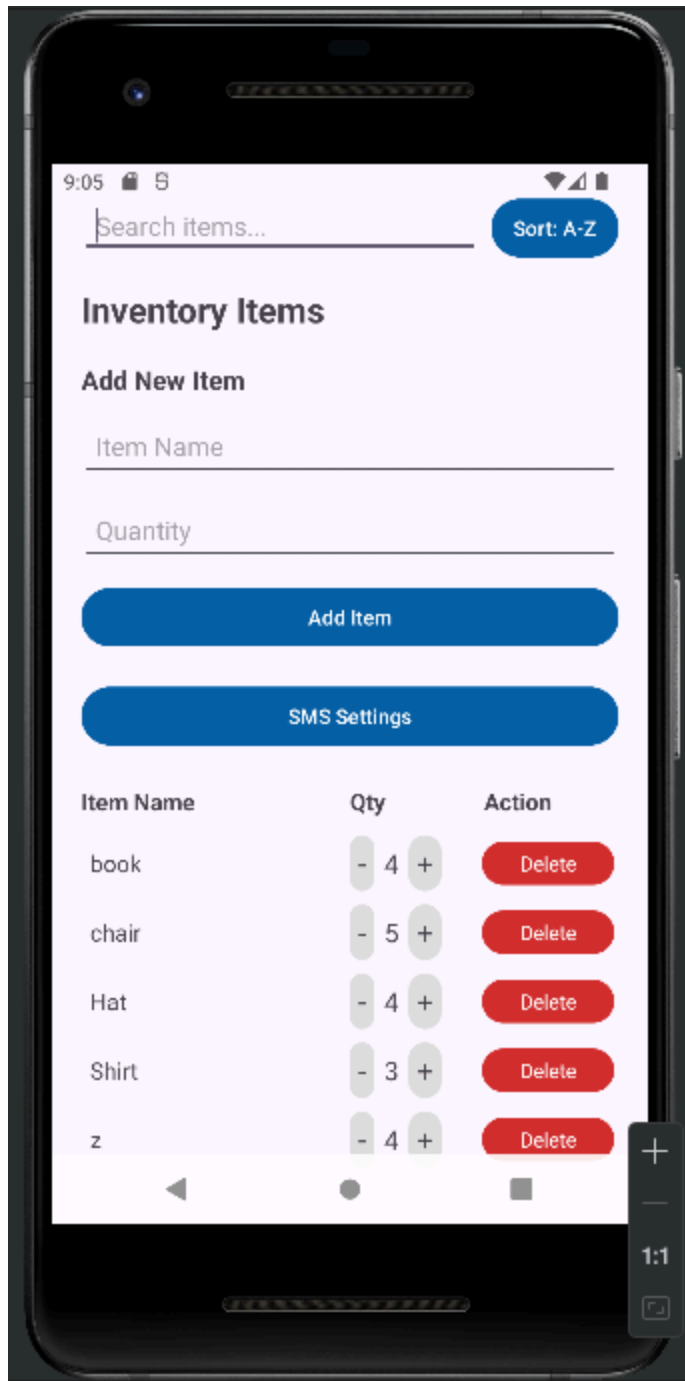
Screenshot 1: Inventory list screen showing the search bar and sort button at the top



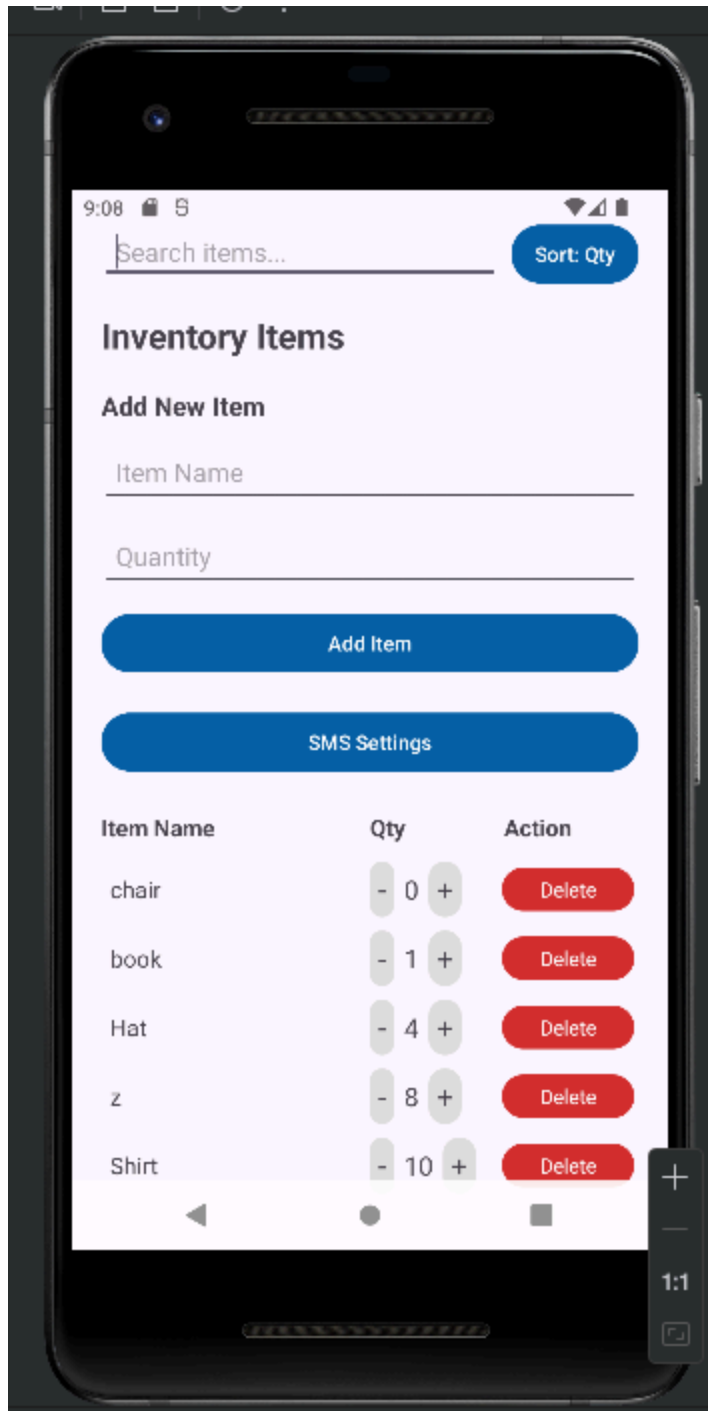
Screenshot 2: Search results filtered by typing a keyword into the search bar



Screenshot 3: Inventory list sorted alphabetically A to Z



Screenshot 4: Inventory list sorted by lowest quantity first



Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome coverage plans?

Yes, this enhancement met the course outcomes planned in Module One. Course Outcome 3 was met because the search and sort feature demonstrates the ability to design a computing solution using algorithmic principles by implementing database level filtering and sorting. Course Outcome 4 was also met because the real time search and sort feature uses industry standard

techniques to deliver clear and measurable usability value to the application. As of right now there are no updates to my outcome coverage plans. The remaining outcomes will be addressed in the final enhancement category.

Reflect on the process of enhancing and modifying the artifact. What did you learn? What challenges did you face?

One thing I learned from this enhancement was how to connect live user input to a database query so the results update in real time. Using a TextWatcher listener to trigger a search query every time the user types a character was something I had not done before and it made the feature feel much more responsive and natural to use.

One challenge I ran into was with the sort order. When I first implemented the A to Z sort, items with spaces or mixed capitalization were not sorting correctly. For example an item called Soccer Ball was appearing after lowercase items instead of in its correct alphabetical position. I fixed this by adding a case insensitive sort directive to the query so that all items sort correctly regardless of how they were named.